Verified Public History Under Physical Time Contracts: Diagnosis, Candidate Coverage, and Stale-Memory Failures

Research draft [1]

Abstract

Robot experience can improve time-constrained recovery, but stronger control candidates and stale history confound this attribution. In a two-process mechanical fixture, we execute 8,640 multi-attempt trajectories, a separately frozen 1,440-attempt reverse-fault extension, and a 720-attempt candidate-complete checkpoint. With the same saved first-attempt history, active public history achieves 66/180 original-contract passes versus 57/180 after context reset: nine paired repairs and no harms. Passive history offers no benefit once candidate coverage is matched. Active history covers exactly the 66 feasible cases of a previously declared, privileged expanded command graph. Under reverse fault changes, the restricted controller harms six cases; two appear temporally successful only after changing the original reference event. These independently audited development results support fault-history adaptation within a finite known model, not neural policy learning, real-time robotics or chemical outcomes.

1. Problem and attribution boundary

An action must complete physically within a window after an event: $T_i - \tau_i \in [L_i, U_i]$. Command acknowledgement is not physical completion. A recovery may consume a shared device or interrupt another process. Success therefore requires all necessary timing, continuity, reference-integrity and eventual task-outcome conditions. A new attempt does not retrospectively repair the failed original attempt.

Our proposed full system turns independently verified success, failure and recovery experience into one shared adaptive policy. A task-level agent generates and diagnoses experience; a digital twin supports calibration and replay; a verifier gates consumption; student-state relabeling tests policy absorption. We distinguish engineering reuse K, public context M and learned parameters θ. This note implements the first two in a mechanical component, with no new trained weights.

GPT-Policy already combines context construction, constrained tools and policy history (Cheng et al., 2026); these alone are not our novelty. SimpleICL studies contextual changes in actions, objects, composition and affordance (Li et al., 2026). AdA studies learned adaptation over changing environments (Adaptive Agent Team, 2023). Here, a known finite fault family gives deterministic support updates. Changing actuator conditions is not held-out chemical task semantics, and support inference is not neural in-context learning.

2. Causal history and physical contracts

At attempt e and local time t, information $\mathcal{F}_e(t)$ contains the contract, public services, observations delivered by global time $s_e + t$, and installation/version-matched prior records. Captured but undelivered observations are excluded. True fault f_e is available only to the plant and independent auditor.

For known noiseless faults $\mathcal{B}_0 = \{\text{effective}, \text{drop first}, \text{persistent}\}$, public action consequences produce compatibility sets $C(z_k)$. Retained support is their intersection, with censored takeover feedback leaving broad support. An eight-pair public response fit defines model support $\mathcal{M}_e$. Within the declared finite candidate family $\mathcal{U}$, admission requires all modeled physical timing and continuity constraints for every $(m, f) \in \mathcal{M}_e \times \mathcal{B}_e$. Independent persisted samples, not these predictions, determine the outcome.

Diagnosis can narrow support while failing the current timing contract. We charge all attempts:

$$J_H = \frac{1}{NH} \sum_{n=1}^{N} \sum_{e=1}^{H} \mathbf{1}\{\text{original contract is TRUE}\}. \quad (1)$$

We also report FALSE, UNKNOWN, diagnostic attempts, command service and paired harms. Clearing context preserves the same first diagnosis. Foreign records retain their

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Table 1. Candidate-complete checkpoint at attempt two. Each arm has 180 cases; all have zero UNKNOWN. Saved first-attempt history is held fixed.

Context	v1 TRUE	v2 TRUE	v2 FALSE
Reset	51	57	123
Passive history	55	57	123
Active history	64	66	114
Foreign rejection	51	57	123

provenance and are rejected; this is an installation-scope control, not a language-order perturbation.

3. Execution and stronger controls

Two driven processes share a serial selective channel and an independent global stop relay. A and B have stop windows; B must remain driven until its minimum duration. Five response settings, three delivery delays, two A windows, two B windows and three faults yield 180 stable cases. Four arms execute ten attempts each (7,200 trajectories). A further 72 configurations execute two arms over ten attempts with a fault change at attempt six (1,440 trajectories). The separately frozen reverse cycle adds 1,440, including first-command-drop to effective transitions.

Every attempt records 1,001 native samples at 10 ms over 10 s. Attempt starts are 11 s apart; pending feedback follows its declared arrival clock, but the gap has no continued dynamics. Host computation does not advance physics. All 10,800 new attempts, including the checkpoint below, retain B continuity; native independent reconstruction error is zero. There is no new RGB input, language-model rollout or real robot. Cases are deterministic development configurations, not independent chemistry batches or population estimates.

The frozen v1 controller gates B-then-relay candidates on fault support and omits relay-only candidates. After finding this coverage confound, we separately freeze a complete-family follow-up with the same saved attempt-two inputs and execute all 720 cases. Fault-independent candidates are now available to all arms. Passive history no longer improves over reset. Active history repairs nine cases and harms none against matched reset (Table 1). This is a single checkpoint, not a replacement ten-attempt curve or free first-attempt diagnosis.

Casewise comparison matches all 66 active passes and all 114 failures to the earlier graph's feasible and infeasible sets. That 10 ms graph preserves original full-drive references, known service and FIFO, but uses privileged fault identity and relaxes feedback waits. The result closes this declared control/information gap; it neither proves a general causal optimum nor provides remaining learned-policy headroom.

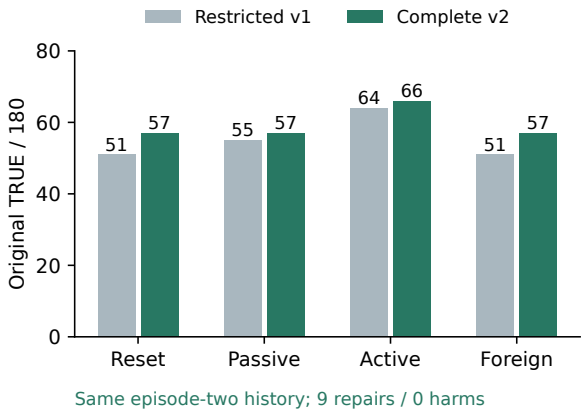


Figure 1. A separately frozen correction to candidate coverage changes the attribution. v1 and v2 share actual prior feedback; no failed cases are removed. The remaining nine-case benefit is deterministic fault-history adaptation, not weight learning.

4. Charged adaptation and stale-history harms

With restricted v1 candidates, stable active history improves from 46/180 at first diagnosis to 64/180 at attempts two through ten. Reset improves from 46 to 51. Across all 1,800 attempts per arm, active history has 622 TRUE, 1,176 FALSE and two UNKNOWN; reset has 505, 1,293 and two. Both charge 180 diagnostic attempts. Their cumulative rates are 34.6% and 28.1%. Passive history gives 546/1,800 without the initial active diagnostic policy. Its difference from active history is not purely memory. These curves do not establish v2 cumulative utility.

At the forward change, active versus reset gives 17 versus 16 passes out of 72, with one repair and no harms. In the reverse extension, the corresponding counts are 11 versus 16, with one repair and six harms. Ten-attempt totals are 194 versus 158 out of 720 in the forward matrix, and 179 versus 158 in the reverse extension. Contradiction resets can recover some cases, but masked consequences need not contradict a stale support, so timely detection is not guaranteed.

A stale belief that the initial command will be dropped can authorize it before the reference. If that command becomes effective, the reference shifts or disappears. All six reverse harms violate independently measured original-reference integrity; two have raw temporal TRUE under the changed reference. We score them FALSE for the original contract. This fixture's full-drive reference is a predeclared semantic requirement; chemistry protocols permitting reference changes need their own definition. The scoring guard detects violations after execution, rather than preventing all such actions.

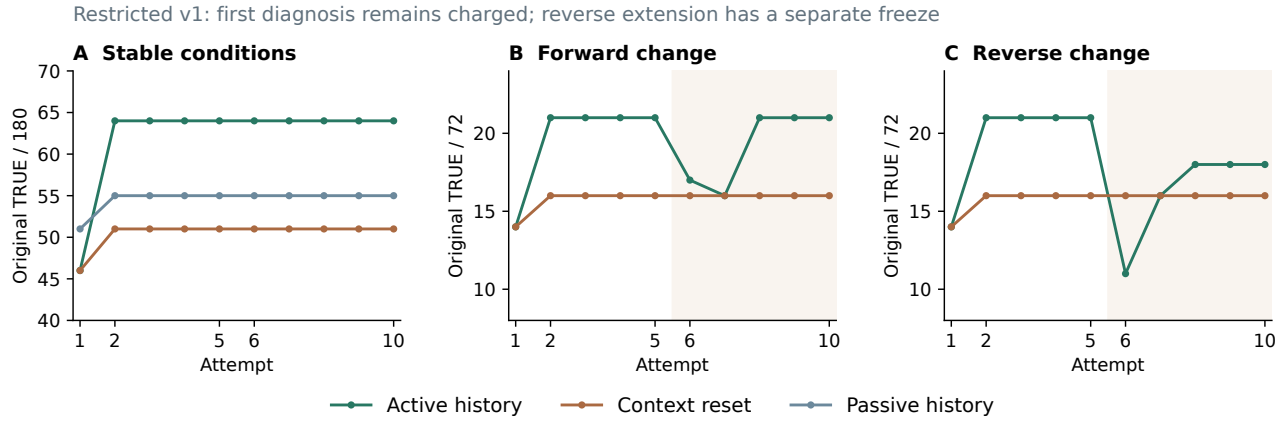


Figure 2. Restricted v1: diagnosis remains charged, and all failed or unknown attempts stay in the denominator. Forward and reverse fault cycles have separate freezes. Foreign-context rejection validates provenance, not VLM sensitivity. The stronger v2 checkpoint is separately reported in Table 1; these curves cannot be extrapolated to v2.

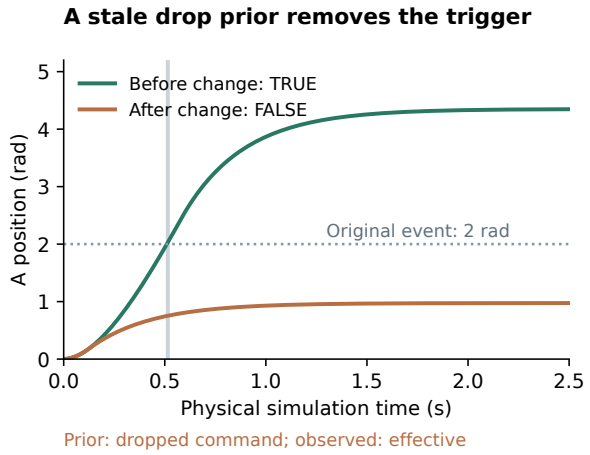


Figure 3. An illustrative audited reverse-cycle installation, selected after complete analysis. Attempt six suppresses the original A position event at 2 rad. Later public evidence resets the prior. This is a negative mechanism example, not an additional test cohort.

5. Verification and next method frontier

Each cohort binds frozen controller and evaluator sources before execution. Independent native integration reconstructs physical effects and FIFO; public controller replay matches all decisions; separately inferred supports check delivery clocks, scope and chronology. An initial pilot audit representation issue and fresh-setting adapter omission are corrected before the main freeze, with failed logs retained and 40 existing pilot trajectories re-audited. No outcome is relabeled by the repair. Raw samples and per-file audit maps remain in the research workspace; the public site provides reports, figures and source-hash summaries, not a full raw-data release.

The complete learned system remains open. Next tests must protect original references under changing faults, then introduce uncertain response models, unseen fault mechanisms and physics that continues during planning. Task-space tests must change goal combinations, dependency graphs or recovery conditions. A shared history/contract-conditioned diffusion policy with student-state expert relabeling and matched DAgger is the initial learning design. The same harness, teacher budget and deterministic candidate-complete control must be held fixed. Disabling the teacher and clearing context distinguishes parameter retention from external help. RL and expert distillation require a demonstrated remaining bottleneck. No claim of neural learning, autonomous RSI, real-time deployment or independent chemical outcomes follows from this component audit.

References

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